



Embedded Software for the Internet Of Things

Design Project - 2025

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Why MSP432?

- Programming MSP432 Launchpad teaches you a lot about the embedded systems and embedded programming as a whole.
 - MSP430 launchpad gives you a better understanding regarding the working of Timers, ADC, DAC, serial communication, interrupts, etc.



Why MSP432?

- You have full control of the system and you can implement **sophisticated** applications.
- Texas Instruments provides solid software development tools:
 - CCS Studio IDE that makes the development of applications as well as their debugging easy.



Other Platforms

- The objective of this course is to teach you the **fundamentals** that can be applied to **any embedded computing platform**.
 - We are not limited to MSP boards!



STM



Arduino



ESP

Other Platforms

- E.g., Arduino is a better choice for absolute beginners and in particular hobbyists.
 - It has a vast community that can support you.
- Using Arduino C, even non-programmers can implement simple applications.



Other Platforms

- However, during our course, we are interested in the **fundamentals**.
- Therefore, we interfaced directly with the hardware, without the support of high-level libraries
- Now, it should look **much much easier** to play with such toy platforms.



Projects

- You can use MSP432 Launchpad and TI Educational Boosterpack Sensors to design your own embedded systems project.
 - You can also select **another hardware platform** and **additional sensors**.
- You can also use Energia (<https://energia.nu/>) to program MSP432 using Arduino C-like high-level language.
 - Even using BoosterPack sensors.



Project Design

- You are going to design and implement a real **embedded/sensing/IoT** application.
- You can also incorporate any other sensors/boards/functionality:
 - E.g., **wireless** communication to make it a real IoT application! -> ESP
 - Other platforms like STM32
- Several websites for purchase online:
 - <https://it.rs-online.com/web/>
 - <https://it.farnell.com/>



STM32 NUCLEO-F401RE



STM32 NUCLEO-F401RE



ESP
(wireless)

Past Projects

- 2024:
 - [Robotic Hand](#)
 - [Voice commands](#)
- 2023:
 - [Claw Machine](#)
 - [Sun Scanner](#)
 - [Oscilloscope](#)
- 2022:
 - [Helicopter Controller](#)
 - [Secure Face](#)
 - [Sensors](#)
 - [Automatic Irrigation System](#)
- 2021:
 - [Receiver & Transmitter on RC-CAR](#)
 - [Automated Car Park Management System](#)
 - [Cube Solver](#)
 - [Planetarium BLE](#)

Past Projects - More

- Domotic House <https://github.com/piedor/domotic-house>
- Small smart home control panel
<https://github.com/StellaDaniele/Embedded-Software-Project>
- SpotifyDirectManager
https://github.com/StefanoDalMas/SpotifyDirectManager_MSP432
- Smart Irrigation System
https://github.com/dongi01/SmartIrrigation_IoTUniTN.git
- PiOS <https://github.com/PiOS-kernel>

Project Groups

- Project group: composed of mostly 3-4 students
- Pair programming as much as possible.
- All members should:
 - put equal effort!
 - have **full control** over the project.
- **Every member of the group is responsible for each part of the system!**



Project Evaluation - Max 15 minutes

- **Maximum 2-minute pitch video** demonstration
- **Maximum 5-minute presentation**
 - Prepare **maximum 5-6 slides**
 - **All** team members will present
 - **Split presentation into equal parts so that everyone presents a part of the project.**
- Approximately 5 minutes Q&A.
- I will ask technical questions to all team members **separately**
 - Where? -> In my room (**Office 120**, Povo 2)

2-Minute Pitch Video

- Problem Statement
 - Describe which problem your system solves.
- Basic Working Scheme
 - Describe how your system works.
 - Which sensors did you use? Explain the basic working flow of the system.
 - If applicable, explain how does the user interacts with your system.

5-MIN. PRESENTATION

- Software Architecture
 - Describe the **core software blocks** of your system.
 - Explain the working flow of the software by giving details on the hardware/software interaction:
 - e.g., which interrupts you used, how you serviced the interrupts? Explain how you sense data, how do you interpret and process the data.
 - Indicate what kind of **data structures** and **algorithms** you used within your code.

5-MIN. PRESENTATION

- **Team Member contributions:**

- If development is separated among team members:
 - **Clearly write who did which part of the project.**
 - Present how many lines of code each member contributed

- **Representative Codes:**

- Present some core C codes for your implementation - > the most important and representative lines (**just a couple of lines**)

5-MIN. PRESENTATION

- Testing
 - Describe how did you test the software.
 - What kind of problems did you encounter? How did you solve them?
- Conclusions and Future Work
 - How would you improve your system?
 - E.g., what additional features and which additional sensors would you add?

PROJECT GRADING

- **Project Evaluation**

- Project demonstration and presentation **%60** -> **all team members**
 - Presentation quality **%10**
 - **Pay attention to address all the sections in the presentation.**
 - Github repository quality **%10**
 - Quality of the project **%40**
 - How sophisticated/interesting the project is (software-wise, algorithms...)
- Technical questions and answers. **%40** -> **individual points**

Course grading (**Tentative**)

%30 Written Exam + **%70** Project Evaluation



minimum 18 to 30



minimum 18 to 30



What to deliver?

- Create a **public** Github Repository
- Your repository should include:
 - **Source code** of your project
 - Including the source code of any other external code
 - **Powerpoint presentation** (or any other format that can be modified)
 - A **Readme** file explaining your project:
 - Check the next slide!

What to deliver? - Readme file

- Your **Readme** file should include the following:
 - What are needed to run the project:
 - Hardware/software requirements
 - Project Layout
 - Source code organization
 - How to build, burn and run your project
 - User Guide
 - Links to your powerpoint presentation and to your Youtube video
 - Team members:
 - explaining clearly how the members contributed to the project
- An example Readme: check [here](#) or other sample projects.